# 1 Linear algebra and set review

**Proof formats (Set inclusion)** Prove $P = Q \implies$ Prove $(P \subseteq Q)$ and $(Q \subseteq P)$. **(Contradiction)** Prove $A \Rightarrow B \implies$ Assume $A$ and $\neg B$ and find contradiction. (**Contrapositive**) Prove $A \Rightarrow B \implies$ Prove $\neg B \Rightarrow \neg A$. **(Induction)** Prove $A(n) \implies$ *Base:* prove $A(0)$, *Assumption:* assume $A(n)$; *Induction:* derive $A(n+1)$.

**Matrix formats** $A = \begin{bmatrix} A_1 & A_2 & \dots & A_n \end{bmatrix} = \begin{bmatrix} a_1^\top & \dots & a_m^\top \end{bmatrix}^\top, \quad Ax = \sum_i A_i x_i = \begin{bmatrix} a_1^\top x & \dots & a_m^\top x \end{bmatrix}^\top$

$A \in \mathbb{R}^{n\times n}$ **invertible** $\iff \operatorname{rank}(A) = n \iff$ linearly independent columns/rows $\iff Ax = b$ unique $x$

**Sherman-Morrison-Woodbury formula** for $A \in \mathbb{R}^{n\times n}, U, V \in \mathbb{R}^{n\times k}$ (special case: $u, v \in \mathbb{R}^n$)

$$(A + UV^\top)^{-1} = A^{-1} - A^{-1}U(I + V^\top A^{-1}U)^{-1}V^\top A^{-1} \qquad (A + uv^\top)^{-1} = A^{-1} - \frac{A^{-1}uv^\top A^{-1}}{(1 + v^\top A^{-1}u)}$$

**Polyhedron** A set $P = \{x \in \mathbb{R}^n \mid Ax \geq b\}$. To **test boundedness** use contradiction: Suppose $\exists d \neq 0$ such that $x + \lambda d \in P \; \forall \lambda \geq 0$, usually arriving at contradiction that $d = 0$.

**Distance** The shortest path from point $x_0$ to plane $a^\top x = b$ is orthogonal to the plane. Write as $x = x_0 - t * a : b = a^\top x = a^\top(x_0 - t * a) = a^\top x_0 - t\|a\|_2^2 \implies t = (b - a^\top x_0)/\|a\|_2^2$.

**Fourier-Motzkin elimination** Tradeoff of decrease in dimension for increase in constraints:

$$P = \{x \in \mathbb{R}^n \mid a_i^\top x \geq b_i, i \in [m]\} = \{x \in \mathbb{R}^n \mid a_{in}x_n \geq -\sum_j a_{ij}x_j + b_i, i \in [m]\} \quad Q = \left\{ x \in \mathbb{R}^{n-1} \mid \begin{cases} d_j + f_j^\top \overline{x} \geq d_i + f_i^\top \overline{x} & a_{jn} < 0, a_{jn} > 0 \\ 0 \geq d_k + f_k^\top \overline{x} & a_{kn} = 0 \end{cases} \right\} = \Pi_{n-1}(P)$$

**Basis** for subset $S$ is a collection $\mathcal{X} = \{x^{(0)}, x^{(1)}, \dots\} \subset S \subseteq \mathbb{R}^n$ such that (i) $\operatorname{span}(\mathcal{X}) = \{y \mid y = \sum_i a_i x^{(i)}\} = S$ and (ii) vectors defining $\mathcal{X}$ are linearly independent.

**Convex function** $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y), \; \forall f : \mathbb{R}^n \mapsto \mathbb{R}, \; \forall x, y \in \mathbb{R}^n, \; \forall \lambda \in [0,1]$

**Convex set** $S \subset \mathbb{R}^n$ such that $\lambda x + (1-\lambda)y \in S, \forall x, y \in S, \lambda \in [0,1]$. A **convex combination** of a finite number of elements of convex set, $S$ also belongs to that set. **(Proof with induction)** Base case: given for 2 elements by definition. Assume step $k$, then

$$\sum_{i=1}^{k+1} \lambda_i x^i = \lambda_{k+1} x^{k+1} + (1-\lambda_{k+1}) \sum_{i=1}^{k} \frac{\lambda_i}{1-\lambda_{k+1}} x^i \qquad \text{with } \lambda_i \geq 0, \sum_i \lambda_i = 1, \; \left( \sum_{i=1}^{k} \frac{\lambda_i}{1-\lambda_{k+1}} x^i \right) \in S$$

**Affine subspace** $y \in S = \{x \mid Ax = b\} \iff (x - x^0) \in S_0 = \{y \mid Ay = 0\}, \; \forall x \in S$ and for fixed $x^0 \in S$. **Dimension.** Let $S = \{x \mid Ax \geq b\}$ and fix any $x^0$ with $Ax^0 \geq b$. Then $S = \{x^0 + y \mid y \in S_0\}, \quad S_0 = \{y \mid Ay \geq 0\}$. The affine hull of $S$ is $\operatorname{aff}(S) = x_0 + \operatorname{span}(x - x_0 : x \in S)$. Its dimension is $\dim(S) = \dim(\operatorname{aff}(S)) = n - \operatorname{rank}(A_{I_=})$, where $I_= = \{i \mid a_i^\top x = b, \forall x \in S\}$.

# 2 Linear programming geometry

**Extreme point** $x \in P$ such that $\nexists y, z \in P, \; \lambda \in [0,1]$ such that $x = \lambda y + (1-\lambda)z$ and $x \neq y, z$. **Vertex** $x \in P$ such that $\exists c$ where $c^\top x < c^\top y, \; \forall y \in P, y \neq x$. **Basic feasible solution** A **basic solution** satisfies $x \in \mathbb{R}^n$ such that i) all equality constraints of $P$ are active, ii) of all active constraints of $P$, $n$ are linearly independent. A **basic feasible solution** is a basic solution with $x \in P$. To **construct a basic solution**, choose $m$ linearly independent columns of $A, A_B$, let $x_j = 0 \; \forall j \notin B$, and solve system of equations, $Ax = b$ for unknowns $x_B = A_B^{-1}b$

**Extreme point $\Rightarrow$ BFS:** Let $I = \{i \mid a_i^\top x = b_i\}$. Suppose $x \in P$ not a BFS, then normals $\{a_i\}_{i\in I}$ do not span $\mathbb{R}^n \implies \exists d \neq 0 \in \mathbb{R}^n$ such that $a_i^\top d = 0 \forall i \in I$. Consider points $y = x + \epsilon d$, $z = x - \epsilon d$. For $i \in I, a_i^\top y = a_i^\top x + \epsilon a_i^\top d = b_i$ so constraints remain active. For $j \notin I$ we have $a_j^\top x > b_j$. Let $\overline{\epsilon} = \min\left\{ \frac{a_j^\top x - b_j}{|a_j^\top d|} \mid j \notin I, \; a_j^\top d \neq 0 \right\} > 0$. Then $\forall \epsilon \in [0, \overline{\epsilon}], \; j \notin I, \; a_j^\top y = a_j^\top x + \epsilon a_j^\top d \geq a_j^\top x - |\epsilon a_j^\top d| > a_j^\top x - (a_j^\top x - b_j) = b_j$. Hence $y, z \in P$. Since $d \neq 0, \; y \neq z, \; x = \frac{1}{2}(y+z)$, a nontrivial convex combination of two distinct feasible points. Contradiction $\square$

**Vertex $\Rightarrow$ Extreme point:** Suppose $x \in P$ a vertex. Then $\exists c$ such that $\forall y, z \in P \setminus \{x\}$, $c^\top x < c^\top z$ and $c^\top x < c^\top y$. This implies $\forall \lambda \in [0,1], \; c^\top x < c^\top(\lambda y + (1-\lambda)z) \implies x \neq \lambda y + (1-\lambda)z \quad \square$

**BFS $\Rightarrow$ Vertex:** Let $x \in P$ a BFS, $I = \{i \mid a_i^\top x = b_i\}$, and $c = \sum_{i\in I} a_i$. Then $c^\top x = \sum_{i\in I} a_i^\top x = \sum_{i\in I} b_i$. Furthermore, $\forall y \in P, c^\top y = \sum_{i\in I} a_i^\top y \geq \sum_{i\in I} b_i$. This shows $x$ optimal to problem of minimizing $c^\top x$. $x$ a BFS $\implies$ there are $n$ linearly independent constraints active at $x$ and $x$ is the unique solution to the system of equations $a_i^\top x = b_i, i \in I \implies x$ is the unique minimizer of $c^\top y, \forall y \in P \implies x$ is a vertex.

**BFS properties and exercises**

- **Finite number of BFS:** For $P = \{x \mid Ax \geq b\}$ with $m$ linearly independent rows of $A$, there are at most $\binom{n}{m}$ BFS
- **BFS implies $x_{j\notin I} = 0$:** Consider $Ax = b, x \geq 0$ with $A \in \mathbb{R}^{m\times n}$ with $m$ linearly independent rows. $x$ a BFS $\iff Ax = b$ and $\exists$ $m$ indices $B$ indices such that $A_B$ are linearly independent. Then $Ax = A_B x_B = b$ and if $j \notin B \implies x_j = 0$.
- **Standard form polyhedra defines $(n-m)$-dimensional affine subspace intersected with the positive orthant.** Let $P = \{x \in \mathbb{R}^3 \mid Ax = b, x \geq 0\}$ and $A \in \mathbb{R}^{m\times n}$ has $m$ linearly independent rows. If $n = m + 1$, then $P$ has at most 2 BFS.
- $\min_{\{x \mid x_1 = 0, x \geq 0\}} x_1$ shows: (i) there may be optimal solutions with support $> m$ (E.g., $(0, x_2, x_3)$), (ii) If there are several optimal solutions, there need not exist two optimal BFS (i.e., only BFS is $(0,0,0)$)
- $\min_{\{x \mid x_1 + x_2 = 1, \; x \geq 0\}} \max\{x_1 - x_2, \; x_2 - x_1\}$ shows: this problem need not have an optimal BFS $\in P$ ($x^* = (\frac{1}{2}, \frac{1}{2})$)

**Existence of extreme points.** For nonempty $P = \{x \mid Ax \geq b\}$ the following are equivalent: (a) $P$ has an extreme point; (b) $P$ contains no line; (c) $A$ contains $n$ linearly independent rows.

- **(b)$\Rightarrow$(a)** Pick any $x \in P$. If $x$ is a BFS, done. Otherwise, let $I = \{i \mid a_i^\top x = b_i\}$ be the active set and note that $A_I$ has linearly dependent rows. Thus $\exists d \neq 0$ with $A_I d = 0$. Since $P$ contains no line, the ray $\{x + \lambda d\}$ cannot stay in $P \; \forall \; \lambda$. Hence at some endpoint $\lambda^*$ of the feasible interval, a new constraint $j \notin I$ becomes tight, $a_j^\top(x + \lambda^* d) = b_j$. Because $a_j^\top d \neq 0$ while $A_I d = 0$, $a_j$ is linearly independent of $\{a_i\}_{i\in I}$. Set $x \leftarrow x + \lambda^* d$ and update $I \leftarrow I \cup \{j\}$. Each iteration strictly increases $rank(A_I)$; after at most $n$ steps we obtain $n$ independent active constraints.



- **(a)$\Rightarrow$(c)** $x \in P$ extreme point $\Rightarrow x$ a BFS $\Rightarrow \exists n$ constraints active at $x \Rightarrow$ corresponding rows of $A$ linearly independent
- **(c)$\Rightarrow$(b)** Contradiction: suppose $\{x + \lambda d : \lambda \in \mathbb{R}\} \subset P$ with $d \neq 0$. Feasibility for all $\lambda$ implies $A(x + \lambda d) \geq b \quad \forall \lambda \implies Ad = 0$. If $A$ has $n$ linearly independent rows, they span $\mathbb{R}^n$ so $Ad = 0 \Rightarrow d = 0$
- **Corollaries.** Bounded polyhedron has an extreme point. $\mathbb{R}^n_{\geq 0}$ contains no line $\implies$ polyhedron in standard form contains no line

**Existence of optimal extreme point.** For $\min_{\{x \mid Ax \geq b\}} c^\top x$, either the cost is $-\infty$ or an optimal extreme point exists. **(Proof)** Let $x \in P$ with active set $I = \{i \mid a_i^\top x = b_i\}$ of rank $k < n \implies \exists d \neq 0$ with $Ad = 0$; choose sign so $c^\top d \leq 0$. Consider $x + \lambda d$. If feasible for all $\lambda > 0$, then $c^\top(x + \lambda d) \to -\infty$. Otherwise let $\lambda^* > 0$ be the first point where a new constraint $j \notin I$ becomes active, $a_j^\top(x + \lambda^* d) = b_j$. Set $x \leftarrow x + \lambda^* d$. Then $c^\top x$ does not increase and $\operatorname{rank}(I)$ increases by 1. Repeat at most $n - k$ times $\square$

**Unique optimum.** For $\min_{\{x \mid Ax = b, x \geq 0\}} c^\top x$ with optimal BFS $x^*$ and reduced costs $\bar{c}_N$, $x^*$ is the unique optimum $\iff I = \{i \in N : \bar{c}_i = 0\} = \emptyset$ (Dual nondegeneracy). **(Proof $\Leftarrow$)** Contradiction: Suppose $\exists y \neq x^*$ with $c^\top y = c^\top x^*$. Let $d = y - x^*$; then $Ad = A_B d_B + A_N d_N = 0$ so $d_B = -A_B^{-1}A_N d_N$. Since $0 = c^\top d = c_B^\top d_B + c_N^\top d_N = \bar{c}_N^\top d_N$. $I = \emptyset \implies \bar{c}_N > 0 \Rightarrow d_N = 0 \Rightarrow d_B = 0$, so $y = x^*$, contradiction. **(Proof $\Rightarrow$)** Contradiction: Suppose $I \neq \emptyset$, and pick $j \in I$ with $\bar{c}_j = 0$. Let $x(\theta) = x^* + \theta d$, $d_N = e_j$, $d_B = -A_B^{-1}A_N e_j$. For small $\theta > 0$, $x(\theta)$ is feasible and $c^\top x(\theta) = c^\top x^* + \theta \bar{c}_j = c^\top x^*$. So a second optimum exists, contradiction

**Full dimensional.** For nonempty $P = \{x \in \mathbb{R}^n \mid Ax \geq b\}$ the following are equivalent: (a) is full dimensional (has positive volume); (b) $\exists \; x \in P : Ax > b \iff$ ; (c) $n+1$ extreme points of $P$ do not lie on a common hyperplane.

- **(a)$\Rightarrow$(c)** Contrapositive: all extreme points must lie on the same hyperplane, $H$, so the convex hull of all extreme points, $P_S$, has dimension $n - 1$. $dim(P) = dim(P_S)$, so $dim(P) = n - 1$.
- **(c)$\Rightarrow$(b)** For each constraint $i$, $\exists \; x^{(i)} : a_i^\top x > b$. Form a convex combination of $x^{(j)} \; \forall j$, which satisfies $x \in P : Ax > b$
- **(b)$\Rightarrow$(a)** Choose interior point $x_0$. Distance from $x_0$ to each constraint is $d_i = (a_i^\top x_0 - b_i)/||a_i||_2$. Note $d_i > 0 \; \forall \; i \implies \epsilon = \min_i d_i > 0$. So ball $B(x_0, \epsilon) = \{y \mid (y - x_0)^\top (y - x_0) \leq \epsilon\} \subseteq P$ has full dimension.

**Formulation: Convert any problem to standard form** WLOG, all rows of $A$ are linearly independent. This can be proved by showing two-way containment between set with and without dependent rows of $A$.

$$\max_{\{x \mid a_i^\top x \geq b_i, a_j^\top x \leq b_j\}} c^\top x \implies \max_{\{x \mid a_i^\top(x^+ - x^-) \geq b_i, a_j^\top(x^+ - x^-) \leq b_j, x^+ \geq 0, x^- \geq 0\}} c^\top(x^+ - x^-) \implies \min_{\{x \mid a_i^\top y - s_i = b_i \; a_j^\top y + s_j = b_j, y, s \geq 0\}} -c^\top y$$

**Maximums and absolute values.** For absolute values, at optimal for each $i$ we must have either $x_i^+ = 0$ or $x_i^- = 0$. Otherwise, we could change both by same amount, preserve feasibility, and reduce cost.

$$\min_{\{x \mid Ax \geq b\}} \max_{i \in [m]} (c_i^\top x + d_i) \implies \min_{\{x \mid Ax \geq b, \; z \geq c_i^\top x + d_i\}} z$$

$$\min_{\{x \mid Ax \geq b\}} \sum_i c_i |x_i| \implies \min_{\{x \mid Ax \geq b, x_i \leq z_i, \; -x_i \leq z_i\}} \sum_i c_i^\top z_i \quad \text{or} \quad \min_{\{x \mid A(x^+ - x^-) \geq b, \; x_i^+, x_i^- \geq 0\}} \sum_i c_i^\top (x_i^+ + x_i^-)$$

# 3 Simplex method

**Basic objects.** Basis $B \in \mathbb{R}^{m\times m}$ (rank $= m$) defines $x_B = A_B^{-1}b$ and $\bar{c}^\top = c^\top - c_B^\top A_B^{-1}A$. Optimal when $x_B \geq 0$ and $\bar{c} \geq 0$.

**Feasible direction.** A vector $d$ is a feasible direction at $x \in P$ if $\exists \theta > 0$ with $x + \theta d \in P$. Feasibility requires

$$A(x + \theta d) = b \implies Ad = 0 \implies A_B d_B + A_N d_N = 0 \implies d_B = -A_B^{-1}A_N d_N.$$

So any feasible direction from a BFS is determined by a choice of $d_N$ (typically $d_N \geq 0$ for moves that keep $x + \theta d \geq 0$).

**Reduced costs & optimality.** The reduced cost vector with basis $B$ is $\bar{c}^\top = c^\top - c_B^\top A_B^{-1}A$. Let $y$ be any feasible point and set $d = y - x$. Then $Ad = 0$, so $d_B = -A_B^{-1}A_N d_N$. The change in objective is

$$c^\top(y - x) = c_B^\top d_B + c_N^\top d_N = c_B^\top(-A_B^{-1}A_N d_N) + c_N^\top d_N = (c_N^\top - c_B^\top A_B^{-1}A_N)d_N = \bar{c}_N^\top d_N.$$

At a BFS, $x_N = 0$, and for any feasible $y$ we have $y_N \geq 0$, hence $d_N = y_N - x_N \geq 0$. If $\bar{c}_N \geq 0 \implies \bar{c}_N^\top d_N \geq 0 \; \forall \; y \in P \implies c^\top y \geq c^\top x \implies x$ is optimal. For optimal $x$ and *infeasible* basic solution, $y$ ($Ay = b, y \not\geq 0$), with $\bar{c}^{(y)} \geq 0$, we have $c^\top y \leq c^\top x$: $c^\top(y - x) = c^\top(y - x) - c_B^\top B^{-1}A(y - x) = \bar{c}^{(y)\top} y - \bar{c}^{(y)\top} x \geq -\bar{c}^{(y)\top} x$ (since $\bar{c}^{(y)} \geq 0$), $-\bar{c}^{(y)\top} x \leq 0$ (since $\bar{c}^{(y)} \geq 0$ and $x \geq 0$).

**Simplex direction.** For entering index $j \in N$: $d_j = 1, \quad d_k = 0 \; (k \in N \setminus \{j\}), \quad d_B = -A_B^{-1}A_j$, since $Ad = 0 \Rightarrow A_B d_B + A_j = 0$

**Simplex step size $\theta$.** We seek the largest $\theta \geq 0$ with $x + \theta d \geq 0$. If $d \geq 0$ then $x + \theta d \geq 0$ for every $\theta \geq 0$ and if $\bar{c}_j < 0$ the objective decreases without bound (unbounded LP). If some $d_i < 0$, feasibility requires $x_i + \theta d_i \geq 0 \Rightarrow \theta \leq -x_i/d_i$. Thus $\theta^* = \min_{i \in B : d_i < 0}\{-x_i/d_i\}$ and the leaving index is the $i$ attaining this minimum. The new BFS is

$$y_j = \theta^*, \qquad y_B = x_B + \theta^* d_B, \qquad y_i = 0 \; (i \in N \setminus \{j\})$$

Whenever $\theta^* > 0$, simplex moves feasible solution and reduces costs ($\theta^* \bar{c}_j < 0$)

**Basis update and independence.** The updated basis $\bar{B} = (B \setminus \{\ell\}) \cup \{j\}$ is linearly independent. If it were dependent, $\exists \lambda \neq 0$ with $A_{\bar{B}}\lambda = 0$. Multiplying by $A_B^{-1}$ yields $A_B^{-1}A_{\bar{B}}\lambda = 0$. But

$$A_B^{-1}A_{\bar{B}} = [e_1, \dots, e_{\ell-1}, \; -A_B^{-1}A_j, \; e_{\ell+1}, \dots, e_m],$$

By construction, the $\ell$th entry is $-d_{B_\ell} \neq 0$ and all other columns are 0 in the $\ell$th entry, thus $A_B^{-1}A_j$ is linearly independent from $A_B^{-1}A_{B\setminus\{\ell\}}$, so $\bar{B}$ is independent and $y$ is a BFS.

**Simplex iteration** (1) Start with BFS $x$ and associated basis $B$. (2) Compute reduced costs $\bar{c}$. If $\bar{c} \geq 0$: $x$ optimal. Else choose entering $j$ with $\bar{c}_j < 0$. (3) Compute $d_B = -A_B^{-1}A_j$. If $d_B \geq 0$: unbounded LP. (4) Compute $\theta^* = \min_{i \in B : d_i < 0} -x_i/d_i$ and leaving index $\ell$. (5) Replace $B_\ell$ with $j$ and compute new BFS $y$.

**Termination (nondegenerate case).** If $\theta^* > 0$ each pivot strictly decreases the objective, $c^\top y = c^\top x + \theta^* \bar{c}_j < c^\top x$. Since finitely many BFS exist and none repeat, simplex terminates.

**Initialization (Phase I).** To find an initial BFS, solve $\min_{\{x,y \mid Ax + y = b, x \geq 0, y \geq 0\}} \mathbf{1}^\top y$. A starting BFS is $(x, y) = (0, b)$. **(Infeasibility)** If the optimum satisfies $\mathbf{1}^\top y > 0$, the original LP is infeasible since this shows $\nexists x \geq 0 : Ax = b$. **(Degeneracy)** If optimal basis has $y = 0$, drop artificial columns and pivot to obtain a pure-$x$ basis. **(Big M)** The Big-$M$ method instead solves min $c^\top x + M\mathbf{1}^\top y$ for sufficiently large $M$.

**Anticycling. (Lexicographic rule)** normalize rows by pivot element; pivot on lexicographically smallest row. Ensures: (i) all rows remain lex-positive, (ii) objective row increases lexicographically, (iii) termination. **(Bland's rule)** choose entering $\min_{\{j:\bar{c}_j<0\}} j$ and leaving $\min_{\{i:d_i<0\}} i$. Prevents cycling.

**Revised simplex.** Naive simplex recomputes $A_B^{-1}$ each iteration ($O(m^3)$). Revised simplex updates inverses via $[A_B^{-1} \mid A_B^{-1}A_j] \xrightarrow{\text{row ops}} [\bar{A}_B^{-1} \mid e_\ell]$ yielding $O(m^2)$ per iteration.

**Upper bound constraints.** $B$ is still the basis of the BFS, but $N$ must be partitioned into $N \supseteq L = \{i : x_i = 0\}$ and $N \supseteq U = \{j : x_j = u_j\}$. Now $\forall\, d_i > 0$, $\theta^*$ will increase $x_i$, possibly over upper bound, $u_i$. Therefore the optimal stepsize determines the exiting variable, which is the arg min of the following

$$\theta^* = \min\left[\min_{\{i|d_i<0, i\in B\}}\left(-\frac{x_i}{d_i}\right),\quad \min_{\{i|d_i>0, i\in B\}}\left(\frac{u_i - x_i}{d_i}\right)\right]$$

Candidate entering variables are $j \in N : (\bar{c}_j < 0, x_j = 0) \cup (\bar{c}_j > 0, x_j = u_j)$.

**Tableau implementation** efficiency comes from maintaining and updating $A_B^{-1}[b \mid A]$. The tableau structure is

$$\left[\begin{array}{c|c} -c_B^\top A_B^{-1} b & c^\top - c_B^\top A_B^{-1} A \\ \hline A_B^{-1} b & A_B^{-1} A \end{array}\right] = \left[\begin{array}{c|ccc} -c^\top x & \bar{c}_1 & \dots & \bar{c}_n \\ \hline x_B & -d^{(1)} & \dots & -d^{(n)} \end{array}\right]$$

Left column: objective value $-c^\top x$ and current basic solution $x_B$. Right block: reduced costs $\bar{c}$ search directions $-d^{(j)} = A_B^{-1}A_j$

| Step | Iteration 1 | Iteration 2 |
|---|---|---|
| 1. Choose pivot column: select $\ell$ with $\bar{c}_\ell < 0$ | $\left[\begin{array}{c\|cccc} 0 & -2 & -1 & 0 & 0 \\ \hline 2 & 1 & -1 & 1 & 0 \\ 6 & 1 & 1 & 0 & 1 \end{array}\right]$ | $\left[\begin{array}{c\|cccc} 6 & -1 & 0 & 0 & 1 \\ \hline 8 & 2 & 0 & 1 & 1 \\ 6 & 1 & 1 & 0 & 1 \end{array}\right]$ |
| 2. Ratio test: among rows, choose $i = \arg\min_{\{d_i^{(\ell)}<0\}}\left(-x_i/d_i^{(\ell)}\right)$ | $\left[\begin{array}{c\|cccc} 0 & -2 & -1 & 0 & 0 \\ \hline 2 & 1 & -1 & 1 & 0 \\ 6 & 1 & 1 & 0 & 1 \end{array}\right]$ | $\left[\begin{array}{c\|cccc} 6 & -1 & 0 & 0 & 1 \\ \hline 8 & 2 & 0 & 1 & 1 \\ 6 & 1 & 1 & 0 & 1 \end{array}\right]$ |
| 3. Apply row operations to get $-d_i^{(\ell)} = 1$ and $d_{k\neq i}^{(\ell)} = 0$ | $\left[\begin{array}{c\|cccc} 0 & -2 & -1 & 0 & 0 \\ \hline 8 & 2 & 0 & 1 & 1 \\ 6 & 1 & 1 & 0 & 1 \end{array}\right]$ | $\left[\begin{array}{c\|cccc} 6 & -1 & 0 & 0 & 1 \\ \hline 4 & 1 & 0 & 1/2 & 1/2 \\ 2 & 0 & 1 & -1/2 & 1/2 \end{array}\right]$ |
| 4. Apply row operations to set $\bar{c}_l = 0$ | $\left[\begin{array}{c\|cccc} 6 & -1 & 0 & 0 & 1 \\ \hline 8 & 2 & 0 & 1 & 1 \\ 6 & 1 & 1 & 0 & 1 \end{array}\right]$ | $\left[\begin{array}{c\|cccc} 10 & 0 & 0 & 1/2 & 3/2 \\ \hline 4 & 1 & 0 & 1/2 & 1/2 \\ 2 & 0 & 1 & -1/2 & 1/2 \end{array}\right]$ |

**Tableau optimality condition exercises.**

$$\left[\begin{array}{cc|cc} -2 & 0 & \alpha \geq 1 & \delta \in [-1,0) \\ \hline 1 & 0 & 3 & 0 \\ \gamma & 1 & -2 & 2 \end{array}\right] \Rightarrow \left[\begin{array}{cc|cc} -2-\gamma\delta/2 & & -\delta/2 & \alpha+\delta \\ \hline 1 & & 0 & 3 \\ & \gamma/2 & 1/2 & -1 \end{array}\right] \Rightarrow \begin{cases} \gamma > 0 & \text{achieve optimal after pivot; original obj was suboptimal} \\ \gamma = 0 & \text{degenerate pivot; original obj was already optimal} \end{cases}$$

$$\left[\begin{array}{cc|cc} * & 0 & 0 & 4 \\ \hline 2 & 1 & 0 & -3 \\ 0 & 0 & 1 & 0 \end{array}\right] \Rightarrow \left[\begin{array}{cc|cc} * & 0 & -\gamma/\alpha & 4 \\ \hline 2 & 1 & 0 & -3 \\ 0 & 0 & 1/\alpha & 0 \end{array}\right] \Rightarrow \begin{cases} \gamma \geq 0 & \text{optimal} \\ \gamma < 0, \alpha > 0 & \text{optimal} \end{cases}$$

# 4 Duality

- Standard form problem ($\min_{\{x|Ax=b, x\geq 0\}} c^\top x$): If $p$ satisfies $p^\top A \leq c^\top \Longrightarrow p^\top b = p^\top Ax \leq c^\top x\ \forall$ feasible $x$.
- General form problem ($\min_{\{x|Ax\geq b\}} c^\top x$): If $p \geq 0$ satisfies $p^\top A = c^\top \Longrightarrow p^\top b \leq p^\top Ax = c^\top x\ \forall$ feasible $x$
- $A_B^{-1}$ uniquely determines $x, p : (x_B, p) = (A_B^{-1}b, c_B^\top A_B^{-1})$: $p^\top b = (c_B^\top A_B^{-1})b = c_B^\top (A_B^{-1}b) = c_B^\top x_B = c^\top x$

**Lagrangian derivation (standard form).**

$$\mathbf{L}(\mathbf{x}, \mathbf{p}) = c^\top x + p^\top(b - Ax) = p^\top b + (c^\top - p^\top A)x \qquad \mathbf{g}(\mathbf{p}) = \min_{x\geq 0} L(x,p) = \begin{cases} p^\top b, & c^\top - p^\top A \geq 0, \\ -\infty, & \text{else} \end{cases} \qquad \max\ \mathbf{g}(\mathbf{p}) = \max_{\{p|p^\top A \leq c^\top\}} p^\top b$$

$(x^*, p^*)$ is primal-dual optimal $\iff$ it is a **saddle point**: $L(x^*, p) \leq L(x^*, p^*) \leq L(x, p^*) \quad \forall x \geq 0,\ p.$

**Sign rules (minimization).**

- **Constraint signs** come from enforcing $L(x,p) \leq c^\top x$ for feasible $x$. For a constraint of the form $Ax \geq b$, feasibility means $b - Ax \leq 0$. Using the Lagrangian term $p^\top(b - Ax)$ with $p \geq 0$ to ensure $L(x,p) = c^\top x + p^\top(b - Ax) \leq c^\top x$, so $g(p) = \inf_x L(x,p)$ is a *lower bound* on the primal optimum.
- **Variable signs** come from making $\inf_x L(x,p)$ finite. If $x \geq 0$, we require $(c^\top - p^\top A) \geq 0$; otherwise the Lagrangian can be pushed to $-\infty$ by sending $x \to +\infty$. If $x$ is free, the only way to avoid $-\infty$ in both directions is $c^\top - p^\top A = 0^\top$

**Common primal-dual pairs**

$$\min\ c^\top x : \{x \mid Ax = b,\ x \geq 0\} \iff \max\ p^\top b : \{p \mid p^\top A \leq c\} \qquad \min\ c^\top x : \{x \mid Ax \geq b\} \iff \max\ p^\top b : \{p \mid p \geq 0,\ p^\top A = c\}$$

$$\min\ c^\top x : \{x \mid Ax \geq b,\ x \geq 0\} \iff \max\ p^\top b : \{p \mid p \geq 0,\ p^\top A \leq c\} \qquad \max\ c^\top x : \{x \mid Ax \leq b,\ x \geq 0\} \iff \min\ p^\top b : \{p \mid p \geq 0,\ p^\top A \geq c^\top\}$$

$$\min\ c^\top x : \{x \mid A_1 x \geq b_1,\ A_2 x = b_2,\ x \geq 0\} \iff \max\ p_1^\top b_1 + p_2^\top b_2 : \{p_1,\ p_2 \mid p_1 \geq 0,\ p_1^\top A_1 + p_2^\top A_2 \leq c\}$$

$$\min\ c^\top x + d^\top y : \{x, y \mid Ax + By \geq e,\ Cx + Dy = f,\ x \geq 0\} \iff \max\ p^\top e + q^\top f : \{p, q \mid p \geq 0,\ p^\top A + q^\top C \leq c^\top,\ p^\top B + q^\top D = d\}$$

**Weak duality.** If $x$ is primal feasible and $p$ is dual feasible, then $p^\top b \leq c^\top x$. **(Proof)** Suppose $x, p$ primal-dual feasible and let $u_i = p_i(a_i^\top x - b_i),\ v_j = (c_j - p^\top A_j)x_j$. By construction, $\text{sign}(a_i^\top x - b) = \text{sign}(p_i)$ and $\text{sign}(c_j - p^\top A_j) = \text{sign}(x_j)$ So $u_i, v_j \geq 0\ \forall i, j$

$$c^\top x - p^\top b = p^\top Ax - p^\top b + c^\top x - p^\top Ax = p^\top(Ax - b) + (c^\top - p^\top A)x = \sum_i u_i + \sum_j v_j \geq 0$$

**Strong duality.** If $(P), (D)$ have optimal solutions, then $c^\top x = p^\top b$. **Simplex proof (standard form).** Let $x^*$ be an optimal BFS with basis $B$. Then $\bar{c}^\top = c^\top - c_B^\top A_B^{-1} A \geq 0$. Define $p^{*\top} = c_B^\top A_B^{-1}$. Then

$$p^{*\top} A = c_B^\top A_B^{-1} A \leq c^\top, \quad p^{*\top} b = c_B^\top A_B^{-1} b = c_B^\top x_B^* = c^\top x^* \qquad \text{(So } p^* \text{ dual feasible and optimal, with equal cost)}$$

**Complementary slackness.** $x, p$ are primal-dual optimal solutions $\iff p \odot (Ax - b) = 0,\ (c - p^\top A) \odot x = 0$. Recall in the proof of Weak duality we defined $u_i, v_j$ and showed $u_i \geq 0,\ v_j \geq 0,\ c^\top x - p^\top b = \sum_i u_i + \sum_j v_j$. **(Proof $\Rightarrow$)** By **Strong duality**, $c^\top x = p^\top b \Longrightarrow \sum_i u_i + \sum_j v_j = 0 \Longrightarrow u_i = v_j = 0\ \forall\, i, j$. **(Proof $\Leftarrow$)** if $u_i = v_j = 0\ \forall\, i, j \Longrightarrow c^\top x - p^\top b = 0 \Longrightarrow c^\top x = p^\top b$.



**Use as an optimality test.** To test candidate $x$ for optimality in $(P)$: (1) Check primal feasibility, (2) Form dual and identify $I = \{i \mid a_i^\top x = b_i\}$, (3) Solve for $p^\top A_I = c_I^\top$, (4) If $p$ satisfies dual feasibility, $x$ is optimal.

**Strict complementary slackness** Given Primal-Dual pair $(P) : \min_{\{x|Ax=b, x\geq 0\}} c^\top x$, $(D) : \max_{\{p|p^\top A\leq c\}} p^\top b$, both with optimal solutions, $\exists$ optimal $x \geq 0, p$ such that $\forall j$ either $x_j > 0$ or $p^\top A_j < c_j$. **(Proof sketch)** (i) first prove that for problem where $x_j = 0$ in all optimal solutions: Set up new primal-dual problems $(P') : \min_{\{x|Ax=b, x\geq 0\}} c^\top x - \delta e_j^\top x$, $(D') : \max_{\{p|p^\top A\leq c-\delta e_j\}} p^\top b$. Consider reduced cost vector $\bar{c}^\top = (c - \delta e_j) - c_B B^{-1} A \Longrightarrow \delta \leq c_j - c_B B^{-1} A_j$. Reason that for sufficiently small $\delta$, same $x^*$ is optimal for $(P')$ and note new constraint of dual implies $p^\top A_j < c_j$. (ii) Then consider the sets of optimal solutions and corresponding sets of indices:

$$\mathcal{X} = \{x \mid Ax = b, x \geq 0, c^\top x = val(P)\}, \quad S = \{j \mid \exists x \in \mathcal{X}, x_j > 0\} \qquad \mathcal{P} = \{p \mid p^\top A \leq c, p^\top b = val(D)\}, \quad S^C = \{j \mid x_j = 0\ \forall x \in \mathcal{X}\}$$

Let $x^{(k)} \in \mathcal{X}$ with $x_k^{(k)} > 0$. Define $\bar{x} = \frac{1}{|S|}\sum_{k\in S} x^{(k)}$, $\bar{p} = \frac{1}{|S^C|}\sum_{k\in S^C} p^{(k)}$, and demonstrate feasiblity and optimality. Now since sets $S, S^C$ partition the index set, we have $\forall j \in [n]$ either $x_j > 0$ or $p^\top A_j < c_j$

**Primal-dual outcome possibilities.** $(P)/(D)$ unbounded $\Rightarrow$ $(D)/(P)$ infeasible. **(Proof)** $(P)$ unbounded $\Rightarrow c^\top x = -\infty \Rightarrow p^\top b \leq -\infty$ by weak duality, which is impossible. $(P)$ finite optimum $\iff$ $(D)$ finite optimum by strong duality. $(P)/(D)$ infeasible $\Rightarrow$ $(D)/(P)$ infeasible *or* unbounded.

**Primal-dual boundedness.** LP $\min_{\{x|Ax\leq b\}} c^\top x$ bounded $\iff$ $c$ is a nonnegative linear combination of the rows of $A$. **(Proof)** Using duality, we see $(D) : \max_{\{p|p\geq 0,\ p^\top A = c\}} p^\top b$ has a feasible set that is precisely this statement. So feasible optimal primal problem $\iff$ bounded optimal primal problem.

**Dual simplex** Helpful to consider in the Tableau structure. Always keep Dual constraints, $\bar{c} = c^\top - p^\top A \geq 0$, and pivot until primal constraints are feasible ($A_B^{-1}b \geq 0$).

$$\left[\begin{array}{c|c} -p^\top b & c^\top - p^\top A \\ \hline A_B^{-1} b & A_B^{-1} A \end{array}\right] = \left[\begin{array}{c|c} -c_B^\top A_B^{-1} b & c^\top - c_B^\top A_B^{-1} A \\ \hline A_B^{-1} b & A_B^{-1} A \end{array}\right] = \left[\begin{array}{c|c} -c^\top x & \bar{c}^\top \\ \hline A_B^{-1} b & A_B^{-1} A \end{array}\right]$$

(1) Start with basic solution $x$ and corresponding basis $B$ with $\bar{c} \geq 0$, (2) choose a row $\ell$ with $v^{(\ell)\top} = (A_B^{-1}b)_\ell < 0$ (if none, terminate with optimal), (3) choose column $j$ with $v_j^{(\ell)} < 0$, (4) use ratio $\gamma^* = \min_{j:v_j^{(\ell)}<0} \frac{\bar{c}_j}{-v_j^{(\ell)}}$ to select the entering variable ($A_j$). If $v^{(\ell)} \geq 0$ terminate as unbounded.

## 4.1 Degeneracy

BFS $x \in P$ is **degenerate** if $> n$ constraints are active at $x$. Degeneracy is representation-invariant for standard-form LPs (changing bases does not change which BFSs degenerate); *not* generally true in non-standard form.

- **Dual nondegeneracy $\Longrightarrow$ unique primal optimum.** See proof in LP geometry section. The converse is not true: (requires redundant constraint): $(P) : \min_{\{x_1=1,\ x_2=1,\ x_1=1,\ x\geq 0\}} 0$, $(D) : \max_{\{p_1+p_3\leq 0,\ p_2\leq 0\}} p_1 + p_2$.
- **Algorithmic consequences.** *Stalling*: new basis but same BFS. *Cycling* return to a previously visited basis.
- **Characterizations.** Standard form (BFS has $m$ basic variables): $x$ degenerate $\iff$ $\exists$ basic variable $x_i = 0$ $\iff$ $|\text{supp}(x)| < m$ $\iff$ same BFS can be obtained from multiple bases. Inequality form: $x$ degenerate $\iff$ $|I(x) = \{i \mid a_i^\top x = b_i\}| > n$ $\iff$ $A_I$ linearly dependent rows.
- **Support size $m$ does not imply BFS.** Counterexample $P = \{x \mid x_1 + x_2 = 2,\ x_3 = 0\}$, $x = [1, 1, 0]$ is feasible and has $|supp(x)| = 2 = m$, but $\text{rank}(A_{\text{supp}(x)}) = 1$
- Degeneracy (primal or dual) means *multiple bases* can represent the same BFS. It does *not* imply multiple optimal solutions.
- Dual basic solution (associated with $B$): $p^\top = c_B^\top A_B^{-1}$, $\bar{c}_N^\top = c_N^\top - p^\top A_N$. Dual BFS is *degenerate* iff some nonbasic variable has reduced cost $\bar{c}_{j\in N} = 0$. More than $n$ inequalities of $p^\top A \leq c$ are tight at the same $p$, so several bases give the same dual BFS $p$.
- Degenerate dual $\not\Rightarrow$ degenerate primal: E.g., $(P) : \min_{\{x|x_1,x_2\geq 0, x_1+x_2=1\}} x_1 + x_2$, $(D) : \max_{\{p|p\leq 1, p\leq 1\}} p$

## 4.2 Farkas' lemma

**Intuition.** Farkas gives a certificate of infeasibility: either (i) $Ax = b,\ x \geq 0$ has a solution, or (ii) $\exists p : p^\top A \geq 0^\top,\quad p^\top b < 0$ that proves no such solution can exist. Suppose true, then $\forall\ x \geq 0,\ p^\top Ax \geq 0$. But since $p^\top b < 0$ we have $p^\top Ax \neq p^\top b \Longrightarrow Ax \neq b$.

**Farkas' theorems of the alternative.** Exactly one holds: $(\exists x \geq 0 :\ Ax = b)$ or $(\exists p :\ p^\top A \geq 0^\top,\ p^\top b < 0)$. Equivalently: $(\forall p : p^\top A \geq 0^\top,\ p^\top b \geq 0) \iff (\exists x \geq 0 :\ Ax = b)$. Cone constraint: $(\exists x \neq 0 :\ Ax = 0, x \geq 0) \iff (\nexists p :\ p^\top A > 0)$. General form inequality (include weak duality): $(\forall x\colon\ Ax \leq b,\ c^\top x \leq d) \iff (\exists p \geq 0 :\ p^\top A = c^\top,\ p^\top b \leq d)$

**Proof using Duality ((a)$\Rightarrow \neg$(b))** $\exists\ x :\ x \geq 0, Ax = b \Rightarrow p^\top b = p^\top Ax$ $\Rightarrow$ so $\nexists p$ such that $p^\top b < 0$ and $p^\top b = p^\top Ax \geq 0$. **($\neg$(a)$\Rightarrow$(b))** Consider primal-dual pair, $(P) : \max_{\{x|x\geq 0, Ax=b\}} 0^\top x$, $(D) : \min_{\{p|p^\top A\geq 0\}} p^\top b$. Not (a) $\Longrightarrow$ $(P)$ infeasible $\Longrightarrow$ $(D)$ unbounded (since $p = 0$ feasible) $\Longrightarrow$ $\exists p$ feasible such that $p^\top b < 0$.

**Proof using Separating hyperplanes ($\neg$(a)$\Rightarrow$(b))** Let $S = \{y \mid x \geq 0,\ Ax = y\} = \{Ax \mid x \geq 0\}$. By Separating Hyperplane theorem, $\exists\ \mathbf{p} : \mathbf{p}^\top \mathbf{b} < \mathbf{p}^\top \mathbf{y}\ \forall\ y \in S$. Now $\forall\ \lambda > 0,\ \lambda A_i \in S \Longrightarrow p^\top b < \lambda p^\top A_i \iff p^\top A > (1/\lambda) p^\top b \Rightarrow \lim_{\lambda\to\infty} p^\top A \geq 0 \Longrightarrow \mathbf{p}^\top \mathbf{A} \geq \mathbf{0}$.

**Farkas' lemma proof examples**

- $(\{x \mid Ax \geq 0,\ x \geq 0\} \subseteq \{x \mid x_1 = 0\})$ $\iff$ $(\exists p \geq 0$ such that $p^\top A \leq 0,\ p^\top A_1 < 0)$: Consider primal-dual pair, $(P) : \min_{\{x|x\geq 0,\ Ax\geq 0\}} -e_1^\top x$, $(D) : \max_{\{p|p^\top A\leq -e_1,\ p\geq 0\}} p^\top 0$. **(Proof $\Rightarrow$)** $val((P)) = 0 \Longrightarrow (D)$ feasible and there exists such a $p$. **(Proof $\Leftarrow$)** $p^\top A_1 < 0 \Rightarrow p^\top A_1 \leq -1 \Rightarrow (D)$ feasible with $val((D) = 0) \Longrightarrow e_1^\top x = 0 \Longrightarrow x_1 = 0$ $\square$
- $(\exists x \neq 0$ such that $Ax = 0, x \geq 0)$ $\iff$ $(\nexists p$ such that $p^\top A > 0)$: **(Proof $\Rightarrow$)** Consider primal-dual pair, $(P) : \min_{\{x|x\geq 0,\ Ax=0\}} 0^\top x$, $(D) : \max_{\{p|p^\top A\leq 0\}} p^\top 0$. $(P)$ feasible $\Rightarrow$ $(D)$ feasible and $p^\top A \leq 0$. **(Proof $\Leftarrow$)** Consider primal-dual pair, $(P) : \min_{\{p|A^\top p\geq e\}} 0^\top p$, $(D) : \max_{\{x|x\geq 0,\ Ax=0\}} e^\top x$ (note $p^\top A < 0 \Rightarrow p^\top A \leq e$ because we can scale). $(P)$ feasible $\Longrightarrow e^\top x = 0$ and $x \geq 0 \Longrightarrow x = 0$ $\square$
- $(\exists x$ with $Ax < 0)$ $\iff$ $(S = \{u \mid u^\top A = 0, u \geq 0, \mathbf{1}^\top u = 1\} = \emptyset)$ **(Proof $\Leftarrow$ using Farkas)** $S = \{u \mid [A, \mathbf{1}]^\top u = [\mathbf{0}^\top, 1]^\top,\ u \geq 0\}$. By Farkas, $S = \emptyset \Longrightarrow \exists\ p, q$ free such that $p^\top A^\top + q\mathbf{1}^\top \geq \mathbf{0}^\top$ and $p^\top \mathbf{0} + q < 0 \Longrightarrow q < 0$ and $Ap > 0 \Longrightarrow Ax < 0$ for $x = -p$. **(Proof $\Leftarrow$ using Strong duality)** Consider primal-dual pair $(P) : \min_{\{u|[A,\mathbf{1}]^\top u=[\mathbf{0}^\top,1]^\top, u\geq 0\}} \mathbf{0}^\top u$, $(D) :$

$\max_{\{(p,q)|Ap+\mathbf{1}q\le 0,(p,q)\text{ free}\}} q$. $(P)$ infeasible $\implies$ $(D)$ unbounded (since $(0,0)$ feasible) $\implies q>0 \implies Ap<0$. **(Proof ⇒ using Strong duality)** Note $Ax<0 \iff Ax\le -\mathbf{1}$. Now consider primal-dual pair $(P):\max_{\{x|Ax\le -\mathbf{1}\}}\mathbf{0}^\top x$, $(D):\min_{\{u|u\ge 0,u^\top A=0\}} -\mathbf{1}^\top u$. $(P)$ feasible $\implies$ $(D)$ feasible $\implies -\mathbf{1}^\top u=0$ by strong duality $\implies S=\emptyset$ since requires $\mathbf{1}^\top u=1$

- Standard form $x_j=0\ \forall x^*$ optimal $\Rightarrow \exists p^*$ such that $p^{*\top}A_j<c_j$ **(Proof)** Define new P-D pair: $(P')\ \min_{\{Ax=b,x\ge 0,\ -c^\top x\ge -f^*\}} -e_j^\top x$, $(D')\ \max_{\{q^\top A-rc^\top\le -e_j^\top,\ q\ge 0\}} q^\top b-rf^*$. Two cases: (i) $r^*>0$: We see that $\frac{q^*}{r^*}$ is optimal and feasible for $(D)$ since $\frac{q^*}{r^*}^\top b=f^*$, $\frac{q^*}{r^*}^\top A\le c^\top-\frac{e_j}{r^*}^\top$. Further, this implies $\frac{q^*}{r^*}^\top A_j<c_j$. (ii) $r^*=0$: Then $q^{*\top}b=0$. For $p^*$ optimal for $(D)$, we see $(p^*+q^*)^\top$ is also optimal and feasible for $(D)$. Further, $(p^*+q^*)^\top A=p^{*\top}A+q^{*\top}A\le c^\top-e_j$ so $(p^*+q^*)^\top A_j<c_j$ $\square$
- $(\exists\ x\ge 0,\ y: Ax+By\ge e,\ Cx+Dy=f) \iff (\nexists\ p\ge 0,\ q: p^\top B+q^\top D=0,\ p^\top A+q^\top C\le 0,\ p^\top e+q^\top f>0)$ **(Proof ⇒)** Contradiction: assume both are true and use all properties to arrive at $p^\top A+q^\top C\not\le 0$

$$p^\top(Ax+By)\ge p^\top e \Longrightarrow p^\top(Ax+By)+q^\top(Cx+Dy)\ge p^\top e+q^\top f>0 \Longrightarrow (p^\top A+q^\top C)x>0 \Longrightarrow p^\top A+q^\top C\not\le 0$$

**(Proof ⇐)** Consider the primal-dual pair below. From this, $(P)$ infeasible $\implies$ $(D)$ unbounded ($(0,0)$ feasible solution) $\implies \exists$ feasible $(p,q): p^\top e+q^\top f\ge 1 \implies p^\top e+q^\top f>0$. $\implies p\ge 0,\ p^\top A+q^\top C\le 0,\ p^\top B+q^\top D=0$

$$(P):\ \min_{\{x,y|Ax+By\ge e,\ Cx+Dy=f,\ x\ge 0\}} x^\top 0+y^\top 0 \qquad (D):\ \max_{\{p,q|p\ge 0,\ p^\top A+q^\top C\le 0,\ p^\top B+q^\top D=0\}} p^\top e+q^\top f$$

### 4.3 Cones and extreme rays

**Definitions (Cone)** A set $C\subset\mathbb{R}^n$ is a cone if $x\in C,\ \lambda\ge 0\Rightarrow\lambda x\in C$. Hence $0\in C$. **(Polyhedral cone)** $\mathcal{C}=\{x\in\mathbb{R}^n\mid Ax\ge 0\}$. **(Recession cone)** For $P=\{x\in\mathbb{R}^n\mid Ax\ge b\}$ and any $y\in P$, $RC_P(y)=\{d\in\mathbb{R}^n\mid y+\lambda d\in P,\ \forall\ \lambda\ge 0\}$. Note $\forall\ y,z\in P$, $RC_P(y)=RC_P(z)=RC_P:=\{d\in\mathbb{R}^n\mid Ad\ge 0\}$. For $Q=\{x\in\mathbb{R}^n\mid Ax=b,\ x\ge 0\}$, $RC_Q=\{d\in\mathbb{R}^n\mid Ad=0,\ d\ge 0\}$.

**Equivalences.** For polyhedral cone $\mathcal{C}=\{x\mid Ax\ge 0\}\subset\mathbb{R}^n$, the following are equivalent: 0 is an extreme point of $\mathcal{C}$ $\iff$ $\mathcal{C}$ does not contain a line $\iff$ $\exists$ $n$ linearly independent rows of $A$. **(Intuition)** A cone contains a line $\iff$ there is a direction $d\ne 0$ with $Ad=0$ and $-d\in C$; equivalently, equalities do not create a BFS at the origin.

**Extreme rays definition.** (1) $x\ne 0\in C\subset\mathbb{R}^n$ is an extreme ray if there are $n-1$ linearly independent constraints active at x. (2) An extreme ray of $RC_P$ with nonempty $P$ is an extreme ray of $P$ (note: $\exists$ finite number of extreme rays in $P$).

**Finite optimal cost over cone.** For $C=\{x\mid Ax\ge 0\}$, $\min_{\{x\in C\}} c^\top x=-\infty \iff \exists$ extreme ray $d\in C$ such that $c^\top d<0$. **(Proof ⇐)** Trivial: If $\exists d\in C$ such that $c^\top d<0$ then $\lambda d\in C\ \forall\ \lambda\ge 0$. **(Proof ⇒)** From given we assume $\exists x\in C$ such that $c^\top x=-1$. Define $P=\{x\mid Ax\ge 0, c^\top x=-1\}\subset C$. $C$ pointed $\Rightarrow$ C does not contain a line $\Rightarrow P$ has at least 1 extreme point, call it $d$. $d$ extreme $\Rightarrow n$ linearly active constraints, one of which must be $c^\top x=-1$. The remaining $n-1$ are from $x\in C\implies d$ extreme ray of $C$ with $c^\top d<0$ $\square$

**Finite optimal cost over polyhedron.** For $P=\{x\mid Ax\ge b\}$, $\min_{\{x\in P\}} c^\top x=-\infty \iff \exists$ extreme ray $d\in RC_P$ such that $c^\top d<0$. **(Proof ⇐)** Trivial: If $\exists d\in RC_P$ such that $c^\top d<0$, then $x+\lambda d\in P,\ \forall\ x\in P,\ \forall\ \lambda\ge 0$. And $c^\top(x+\lambda d)=c^\top x+\lambda c^\top d\to-\infty$. As $\lambda\to\infty$. **(Proof ⇒)** Consider primal-dual pair $(P):\min_{\{x\in P\}} c^\top x$, $(D):\max_{\{p\in Q=\{p|p\ge 0,p^\top A=c\}\}} p^\top b$. $(P)$ unbounded $\implies$ $(D)$ infeasible $\implies$ $(D'):\max_{\{p\in Q\}} p^\top\mathbf{0}$ infeasible. Dual of $(D')$, $(P')=\min_{\{x\in\tilde P=\{x|Ax\ge 0\}\}} c^\top x$ is unbounded (since $x=0$ feasible) $\Longrightarrow\exists$ extreme ray $d\in\tilde P$ such that $c^\top d<0$ (by previous theorem). $\tilde P=RC_P\Longrightarrow$ extreme ray of $RC_P$ is extreme ray of $P$ $\square$

**Resolution theorem** For $P=\{x\mid Ax\ge b\}$ nonempty, $\{x^{(k)}\}_{k\in\mathcal{K}_P}\ne\emptyset$ all its BFS, $\{w^{(r)}\}_{r\in\mathcal{R}_P}$ all its extreme rays. Then $P=Q=\left\{\sum_{k\in\mathcal{K}_P}\lambda_k x^{(k)}+\sum_{r\in\mathcal{R}_P}\theta_r w^{(r)}\mid\lambda_k\ge 0,\theta_r\ge 0,\sum_{k\in\mathcal{K}_P}\lambda_k=1\right\}$. **(Proof $Q\subset P$)** Trivial. Evaluate $x\in Q$ for $P$ feasibility. **(Proof $P\subset Q$)** Contradiction: Suppose $\exists z\in P,\ z\notin Q$. Consider primal-dual pair $(P):\max_{\{(\lambda,\theta)|z=X\lambda+W\theta,\ \mathbf{1}^\top\lambda=1,\ \lambda,\ \theta\ge 0\}}\mathbf{0}$, $(D):\min_{\{(p,q)|p^\top X+q\mathbf{1}^\top\ge 0,p^\top W\ge 0\}} p^\top z+q$. By assumption ($z\notin Q$), $(P)$ infeasible $\implies$ $(D)$ unbounded (since $(p,q)=(0,0)$ is feasible) $\Longrightarrow\exists(p,q)$ such that $p^\top z+q<0$ and $p^\top X\ge -q\mathbf{1}^\top$, $p^\top W\ge 0^\top\Longrightarrow p^\top z<p^\top x^{(k)},\ \forall\ k\in\mathcal{K}_P$. Now consider new problem $(P'):\min_{\{Ax\ge b\}} p^\top x$ which has maximum at $x^{(k^*)}$ for some $k^*\in\mathcal{K}_P$. Since $z\in P$ is feasible this means $p^\top z\ge p^\top x^{(k^*)}$, arriving at our contradiction to $p^\top z<p^\top x^{(k)},\ \forall\ k\in\mathcal{K}_P$ $\square$

## 5 Sensitivity analysis

Given starting optimal basis, $B$, and associated optimal solution, $x^*$, we have $B^{-1}b\ge 0$ (feasibility) and $\bar c=c-c_B^\top B^{-1}A\ge 0$ (optimality). This section examines how these change with changes to $(A,b,c)$ and how to efficiently solve for this deviation.

**New variable.** $\min_{\{x|x\ge 0,Ax+A_{n+1}x_{n+1}=b\}} c^\top x+c_{n+1}x_{n+1}$. Note that $(x,x_{n+1})=(x^*,0)$ is a BFS to this new problem under original $B$. Suffice to check optimality of $\bar c_{n+1}=c_{n+1}-c_B^\top B^{-1}A_{n+1}\ge 0$. To solve: augment the simplex tableau with an additional column, $A_{n+1}$ and continue iterations.

**New inequality.** Introduce constraint $a_{m+1}^\top x\ge b_{m+1}$. If $x^*$ satisfies this constraint, the augmented problem is optimal. Else, bring into standard form with new variable, $x_n+1$ by defining

$$\overline{A}=\begin{bmatrix}A & 0\\ a_{m+1}^\top & -1\end{bmatrix}\quad \overline{B}=\begin{bmatrix}B & 0\\ a_{m+1_B}^\top & -1\end{bmatrix}\quad \overline{B}^{-1}=\begin{bmatrix}B^{-1} & 0\\ -a_{m+1_B}^\top B^{-1} & -1\end{bmatrix}$$

To solve: observe that this problem is dual-feasible (infeasible x, but nonnegative reduced cost vector), so use the dual simplex beginning at current BFS.

**Change in $b$.** Suppose $b'\leftarrow b+\delta e_i$. Optimality condition ($\bar c\ge 0$) unaffected, only feasibility ($x'_B=B^{-1}b'\ge 0$) must be considered. We have $B^{-1}(b+\delta e_i)=x_B+\delta B_i^{-1}$. For any $\delta$ in

$$\max_{\{j|[B^{-1}]_{ji}>0\}}\left(-\frac{x_{B(j)}}{[B^{-1}]_{ji}}\right)\le\delta\le\min_{\{j|[B^{-1}]_{ji}<0\}}\left(-\frac{x_{B(j)}}{[B^{-1}]_{ji}}\right)$$

optimality and primal feasiblity are satisfied. For $\delta$ outside this range, optimality (dual feasibility) is still satisfied, but not primal feasibility. Use dual simplex to find new solution.

**Change in $c$.** Suppose $c'\leftarrow c+\delta e_j$. Feasibility condition ($x_B=B^{-1}b\ge 0$) unaffected, only optimality ($\bar c'=c'-c_B'^\top B^{-1}A\ge 0$) must be considered. If $j$ nonbasic, suffice to check $c_B^\top B^{-1}A_j\le c_j+\delta \iff \bar c_j\ge -\delta$. If $j=\ell\in B$, $\bar c'^\top=(c+\delta e_j)^\top-(c_B+\delta e_\ell)^\top B^{-1}A$

$$\bar c_i':\quad \bar c_i=0\ (i\in B\setminus\ell)\qquad \bar c_i+\delta-\delta=\bar c_i=0\ (i=\ell\in B)\qquad \bar c_i-\delta e_\ell^\top B^{-1}A_i\ (i\in N)$$

Suffice to check $\bar c_N'^\top=\bar c_N^\top-\delta e_\ell^\top B^{-1}A_N\ge 0\Longrightarrow\bar c_N\ge\delta e_\ell^\top B^{-1}A_N$. In both cases, if true, current solution is still optimal, else apply primal simplex starting at the current BFS.



**Change in nonbasic column of $A$.** Suppose $a'_{ij}\leftarrow a_{ij}+\delta$. If $A_j$ nonbasic, then $B$ is unaffected and hence primal feasiblity is unaffected. Only reduced cost of column $j$ is affected. Condition to check is $c_j-c_B^\top B^{-1}(A_j+\delta e_i)\ge 0 \iff \bar c_j-\delta p_i\ge 0$. If this condition is violated, bring nonbasic column $A_j$ into the basis and contine with the primal simplex method.

**Change in basic column of $A$.** Suppose $a'_{ij}\leftarrow a_{ij}+\delta$ and $A_j$ basic with $j=B(\ell)$. Then $B'\leftarrow B+\delta e_i e_\ell^\top$. Must reason about feasibility ($x'_B=(B+\delta e_i e_\ell^\top)^{-1}b$) and optimality ($\bar c'^\top=[\bar c_B'^\top,\bar c_N'^\top]=[0,c_N^\top-c_B(B+\delta e_i e_\ell^\top)^{-1}N)$. Must reason about inverse (use Sherman-Morrison formula) to bound $\delta$.

**Global sensitivity** $G(c)=\min_{\{Ax=b,x\ge 0\}} c^\top x=\min_{\{i\in[N]\}} c^\top x^{(i)}$, a convex piecewise linear function of c. Use the primal problem to study this since changing $c$ affects optimality and not feasibility. $F(b)=\min_{\{Ax=b,x\ge 0\}} c^\top x=\max_{\{j\in[M]\}} p^{(j)\top}b$, a convex piecewise linear function of b. Use dual problem to study since changing $b$ affects dual optimality and not dual feasibility.

## 6 Large scale optimization

**Delayed column (row) generation.** Column generation either confirms we're optimal ($\min_{j\notin I}\bar c_j\ge 0$) or provides new *variable* $j$. Row generation either confirms we're feasible ($z\ge\max_{j\notin I}\{a_j^\top x-b_j\}$) or provides new *constraint* $j$.

$$\min\ c^\top x\text{ s.t. }Ax=b,\ x\ge 0\Longrightarrow\min\ c_I^\top x_I\text{ s.t. }A_I x_I=b,\ x_I\ge 0\Longrightarrow\min_{j\notin I}\bar c_j=c_j-p^\top A_j \qquad\text{(column generation)}$$

$$\min\ z\text{ s.t. }z\odot e\ge Ax-b\Longrightarrow\min\ z\text{ s.t. }z\odot e\ge A_I x-b_I\Longrightarrow\max_{j\notin I}\ a_j^\top x-b_j \qquad\text{(row generation)}$$

### 6.1 Dantzig-Wolfe decomposition.

$\min_{x_1,x_2} c_1^\top x_1+c_2^\top x_2$ s.t. $D_1x_1+D_2x_2=b_0,\ x_1\in P_1,\ x_2\in P_2$, $P_i=\{x_i\ge 0\mid F_ix_i=b_i\}$ (Too many variables, partition feasible sets into blocks, $P_i$)

**Master problem.** Reformulate original problem using resolution theorem:

$$\min_{\lambda,\theta}\sum_{i\in\{1,2\}}\left(\sum_{j\in\mathcal{J}_i}\lambda_i^j c_i^\top x_i^j+\sum_{k\in\mathcal{K}_i}\theta_i^k c_i^\top w_i^k\right)\ \text{ s.t. }\sum_{i\in\{1,2\}}\left(\sum_{j\in\mathcal{J}_i}\lambda_i^j D_i x_i^j+\sum_{k\in\mathcal{K}_i}\theta_i^k D_i w_i^k\right)=b_0,\ \mathbf{1}^\top\lambda_i=1,\ \lambda_i^j,\theta_i^k\ge 0,\ \forall i,j,k$$

**Restricted master.** Solve using only a subset of $\{x_i^j,w_i^k\}$. Let duals be $q^k$ for $D_1x_1+D_2x_2=b_0$ and $r_i^k$ for $\mathbf{1}^\top\lambda_i=1$.

**Subproblem.** Form $\min_{\{x_i|x_i\in P_i\}}(c_i^\top-q^{k\top}D_i)x_i-r_i^k$: **(i) Unbounded**: $\exists w_i^k$ with $\bar c=(c_i^\top-q^\top D_i)w_i^k<0\Rightarrow$ add variable $\theta_i^k$ and parameters $(c_i^\top w_i^k,\ D_iw_i^k)$. **(ii) Finite and $<r_i^k$**: $\exists x_i^j$ with $\bar c=(c_i^\top-q^\top D_i)x_i^j-r_i<0\Rightarrow$ add variable $\lambda_i^j$ and parameters $(c_i^\top x_i^j,\ D_ix_i^j)$.**(iii) Finite and $\ge r_i^k$ for all $i$**: $\bar c\ge 0\Rightarrow$ optimal

**Iteration bound** The dual function is a lower bound on our master problem. At iteration $k$, we have dual-feasible values $q^k$ and $\min_{x_i\in P_i}(c_i^\top-q^{k\top}D_i)x_i$. Together these can be used to construct a lower bound:

$$\min_{\lambda,\theta}\ L(\lambda,\theta,q\ge 0,r)=\min_{\lambda,\theta}\left\{c^\top(X\lambda+W\theta)+q^\top(b_0-D(X\lambda+W\theta))+\sum_{i=1}^2 r_i(1-e^\top\lambda_i)\right\} \qquad\text{(Lagrangian)}$$

$$g(q\ge 0,r)=q^\top b_0+\sum_{i=1}^2\left(r_i+\min\left\{\min_{\lambda_i}(c_i^\top X_i-q^\top D_iX_i-r_ie^\top)\lambda_i,\ \min_{\theta_i}(c_i^\top W_i-q^\top D_iW_i)\theta_i\right\}\right) \qquad\text{(Dual problem)}$$

$$=q^\top b_0+\sum_{i=1}^2\left(\min\left\{\min_{x_i^j}(c_i^\top x_i^j-q^\top D_ix_i^j),\ \min_{w_i^k}(c_i^\top w_i^k-q^\top D_iw_i^k)\theta\right\}\right)=q^\top b_0+\sum_{i=1}^2\min_{x_i\in P_i}(c_i^\top-q^\top D_i)x_i$$

(optimal at extreme point or ray)

For any basic $\lambda_i^j:\bar c_i=(c_i^\top-q^\top D_i)x_i^j-r_i=0\Longrightarrow r_i=(c_i^\top-q^\top D_i)x_i^j\ge\min_{x_i\in P_i}(c_i^\top-q^\top D_i)x_i=:z_i$

### 6.2 Benders decomposition

$\min_{x,y} c^\top x+f^\top y$ s.t. $Ax=b,\ Dx+By=d,\ x,y\ge 0$ (Too many constraints, keep $x$ in the master, eliminate $y$ via a subproblem)

**Master problem.** Eliminate $y$ via its subproblem and represent its cost by $z$. This significantly reduces the number of variables, but increases the number of constraints. Consider $x$ fixed, let $p$ be the dual variable for $By=d-Dx$, and let $\mathcal{P}=\{p\mid p^\top B\le f^\top\}$. Denote the extreme points of $\mathcal{P}$ by $p^i$ and the extreme rays by $w^j$. The full reformulation of the original problem is

$$\min_{x,z} c^\top x+z\quad\text{s.t. }Ax=b,\ x\ge 0,\ p^{i\top}(d-Dx)\le z,\ w^{j\top}(d-Dx)\le 0,\quad\forall\text{ extreme points }p^i\in\mathcal{P},\text{ extreme rays }w^j\in\mathcal{P}$$

**Restricted master.** Solve using the same objective as the master problem, but only a subset of $\{p^i,w^j\}$. For a given optimal solution to the restricted problem, $(x^k,z^k)$, check feasibility in master plrobblem.

**Subproblem.** Form $(P)\ \min_{\{y|By=d-Dx^k,y\ge 0\}} f^\top y$ and $(D)\ \max_{\{p|p^\top B\le f^\top\}} p^\top(d-Dx^k)$: **(i)** $(D)$ **unbounded**: $\exists\ w^j$ with $w^{j\top}(d-Dx^*)>0\Rightarrow$ add constraint $w^{j\top}(d-Dx)\le 0$ to restricted problem **(ii)** $(P)$ **finite and $<z^k$**: $\exists\ p^i$ with $p^{i\top}(d-Dx^*)>z^*\Rightarrow$ add constraint $p^{i\top}(d-Dx)\le z$ to restricted problem **(iii)** $(P)$ **finite and $\ge z^k$**: optimal solution to restricted problem that is feasible for the master problem $\Rightarrow$ optimal

**Integer knapsack problem.** Solved as a dynamic programming problem

$$F(u)=\max_a\sum_{i=1}^m p_ia_i\text{ s.t. }\sum_{i=1}^m w_ia_i\le u,\ a\in\mathbb{Z}_+^m \qquad F(u)=\begin{cases}0 & u<\min_i w_i\\ \max_{i\in[m]}\{p_i+F(u-w_i)\} & \text{else}\end{cases}$$

## 7 Network flows

**A tree** is an undirected graph with no cycles. Properties: i) An undirected graph is a tree $\iff$ it is connected and has $|\mathcal{N}|-1$ arcs. ii) For any distinct nodes $i,j$ in a tree, $\exists$ a unique path from $i\leftrightarrow j$. *Proofs involve deleting leaves and arcs in cycles*

**A spanning tree** of an undirected graph $G=(\mathcal{N},\mathcal{E})$ is a tree, $T=(\mathcal{N},\mathcal{E}_1)$, such that $\mathcal{E}_1\subseteq\mathcal{E}$. Properties: i) Let $\mathcal{E}_0\subseteq\mathcal{E}$ without any cycles $\Longrightarrow\mathcal{E}_0$ can be augmented to $\mathcal{E}_1\supseteq\mathcal{E}_0$ so that $(\mathcal{N},\mathcal{E}_1)$ is a spanning tree

**Network flow.** Directed graph, $G=(\mathcal{N},\mathcal{A})$ with nodal supply $b_i$, arc capacity $u_{ij}$, and flow cost $c_{ij}$

$$\min_{f_{ij}} \sum_{(i,j)\in\mathcal{A}} c_{ij}f_{ij}, \quad \text{s.t. } b_i + \sum_{j\in I(i)} f_{ji} = \sum_{j\in O(i)} f_{ij}\ \forall\, i\in\mathcal{N}, \quad 0\le f_{ij}\le u_{ij}\ \forall\,(i,j)\in\mathcal{A}$$

Every network flow problem can be reduced to i) one with exactly one source and one sink [new outgoing (incoming) node connected to each source (sink); zero cost arcs; arc capacity equal to nodal supply], ii) one without sources or sinks [connect $t\to s$ with $u_{ts}=b_s,\ c_{ts}=-M$]

**Transportation problem** (**Assignment problem** sets $s_i=1,\ d_j=1,\ |\mathcal{N}_d|=|\mathcal{N}_s|$)

$$\min_{f_{ij}} \sum_{i=1}^{m}\sum_{j=1}^{n} c_{ij}f_{ij}, \quad \text{s.t. } \sum_{i=1}^{m} f_{ij}=d_j\ \forall\, j\in\mathcal{N}_d, \quad \sum_{j=1}^{j} f_{ij}=s_i\ \forall\, i\in\mathcal{N}_s, \quad f_{ij}\ge 0\ \forall\,(i,j)\in\mathcal{A}$$

**Shortest path.** Minimum length path to target node $t$ from source node(s) $s\in S\subseteq\mathcal{N}\setminus\{t\}$. Formulated as a network flow problem by redefining $b_i$. Dual lends itself to Bellman equation and Dijkstra's algorithm.

$$b_i=\begin{cases}|S| & i=t\\ +1 & i\in S\\ 0 & \text{else}\end{cases} \qquad (D):\max_{\{p_i\le c_{ij}+p_j,\ \forall\ (i,j)\in\mathcal{A}\}} \sum_{i\in\mathcal{N}} b_ip_i \qquad (D'):\max_{\{p_i\le c_{ij}+p_j,\ \forall\ (i,j)\in\mathcal{A},\ S=\mathcal{N}\setminus\{t\},\ p_t=0\}} \sum_{i\neq t} p_i$$

Properties: i) if $\exists$ negative length cycle, optimal cost is $-\infty$. ii) if all cycles have nonnegative length and a feasible tree solution is optimal, then the tree is a tree of shortest paths. ii) if all cycles have nonnegative length and we fix $p_n=0$, then $p^*$ unique and $p_i^*$ is shortest path length from node $i\to t$.

**Maximum flow.** Largest possible flow sent through the network from $s\to t$.

$$\max\ b_s \quad \text{s.t. } Af=b,\ b_t=-b_s,\ b_i=0\ \forall\, i\notin\{s,t\},\ 0\le f_{ij}\le u_{ij}\ \forall\,(i,j)\in\mathcal{A}$$

Reformulate as network flow: $\mathcal{A}'=\mathcal{A}\cup\{(t,s)\},\ c_{ij}=\begin{cases}-1 & (i,j)=(t,s)\\ 0 & \text{else}\end{cases},\ u_{ts}=\infty,\ b_i=0$

Dual: $\min \sum_{(i,j)\in\mathcal{A}} u_{ij}\theta_{ij}$ s.t. $p_i-p_j+\theta_{ij}\ge 0\ \forall\,(i,j)\in\mathcal{A},\ p_t-p_s\ge 1,\ \theta_{ij}\ge 1\ \forall\,(i,j)\in\mathcal{A}$

**Minimum cut.** Global cut problem creates $S\subset\mathcal{N}$ such that minimum capacity passes from $S\to S^C$. The value of the cut is $C(S)=\sum_{\{(i,j)\in\delta(S)\}} u_{ij}$, $\delta(S)=\{(i,j)\mid i\in S,\ j\notin S\}$. Add constraint $(x_s,x_t)=(1,0)$ to make Minimum s-t cut problem. Can be proved the $s\to t$ cut is the dual of the $s\to t$ Max Flow problem

$$\min_{\emptyset\subset S\subset\mathcal{N}} \sum_{(i,j)\in\delta(S)} u_{ij} \implies \min_{x,z} \sum_{(i,j)\in\mathcal{A}} u_{ij}z_{ij} \quad \text{s.t. } z_{ij}\ge x_i-x_j\ \forall\,(i,j)\in\mathcal{A},\ 0\le x_i,z_{ij}\le 1 \qquad x_i=\mathbb{I}\{i\in S\},\ z_{ij}=\mathbb{I}\{i\in S,\ j\notin S\}$$

**Matrix notation.** $A: a_{ik}=1$ $(-1)$ if $i$ starts (ends) arc $k$ and $a_{ik}=0$ otherwise. When $Af=0$ we have a **circulation**. The rows of $A$ are not linearly independent, so we define $\tilde{A}=A_{[:n-1,:]}$ $\tilde{A}$ has linearly independent rows if $G$ connected **(Proof)** $G$ connected $\implies\exists$ spanning tree $T\subset\mathcal{A}$. Form $\tilde{A}_T$ and use unique tree solution result (below) to show $\tilde{A}_T$ singular.

**A tree solution** is a flow vector, $f$, constructed by (a) picking $T\subset A$ of $n-1$ arcs forming a *spanning* tree when direction is ignored, (b) setting $f_{ij}=0\ \forall\,(i,j)\notin T$, (c) solving $\tilde{A}\tilde{f}=\tilde{b}$ to determine $f_{ij}\ \forall\,(i,j)\in T$. **A feasible tree solution** is a tree solution satisfying $f\ge 0$ (or $d\preceq f\preceq u$ if capacitated). **A tree solution is a unique solution: (Proof sketch)** suffice to show that $\tilde{A}_T$ is nonsingular: reorder its rows and columns to show $\tilde{A}_T$ is lower triangular with nonzero diagonals, hence nonsingular.

**A flow vector is a basic solution $\iff$ it is a tree solution. (Proof $\Leftarrow$)** follows immediately from above **(Proof $\Rightarrow$)** Contrapositive: Suppose not $f$ not a tree solution and consider case when $Af=b$ (if not equal, not basic by definition). Let $S=\{(i,j)\in\mathcal{A}\mid f_{ij}\neq 0\}$. If no cycles, $\exists T\supset S$ forming a spanning tree and $f$ is the tree solution associated with $T$, contradiction. If cycles, let $h^C$ be a circulation and we have $A(f+h^C)=b$ and we observe whenever $f_{ij}=0\Rightarrow h^C_{ij}=0$ and constraints active at $f$ and $f+h^C$, so no unique solution and $f$ not basic.

**Simplex change of basis and reduced costs.** For spanning tree, $T$, pick arc $(i,j)\notin T$ and identify cycle that is formed. Let $F,B$ be forward and backward arcs in cycle relative to $(i,j)$ forward. In capacitated case, assume $d_{ij}\le f_{ij}\le u_{ij}$.

$$\hat{f}_{k\ell}=\begin{cases}f_{k\ell}+\theta & \text{if } (k,\ell)\in F\\ f_{k\ell}-\theta & \text{if } (k,\ell)\in B\\ f_{k\ell} & \text{otherwise}\end{cases} \quad \theta^*=\begin{cases}\text{(uncap.)} & \min_{(k,\ell)\in B} f_{k\ell}\\ \text{(cap.)} & \min\{\min_{(k,\ell)\in B} f_{k\ell}-d_{k\ell},\min_{(k,\ell)\in F} u_{k\ell}-f_{k\ell}\}\end{cases} \quad \bar{c}_{ij}=\left(\sum_{(k,\ell)\in F} c_{k\ell}-\sum_{(k,\ell)\in B} c_{k\ell}\right)=c_{ij}-(p_i-p_j)$$

For $p^\top=c_T^\top\tilde{A}_T^{-1}$, $p_n=0$. Since $\bar{c}_T=0$, we solve for $p$ given the system of equations $p_i-p_j=c_{ij}\ \forall\,(i,j)\in T, p_n=0$. Without fixing $p_n$, we can add a constant to all $p$ without changing the results.

**Network simplex iteration.** (1) Start with BFS flow $f$ and spanning tree $T$ (add artificial arcs with cost M if needed). For capacitated problems, define D (at lower bound) and U (at upper bound). (2) Compute node potentials $p$ by fixing $p_n=0$ and enforcing $p_i-p_j=c_ij\ \forall\,(i,j)\in T$. (3) Compute reduced costs $\bar{c}_{ij}=c_{ij}-(p_i-p_j)\ \forall\,(i,j)\notin T$. If $\bar{c}\ge 0\Rightarrow x$ optimal and terminate. Else choose $(i,j):\bar{c}_{ij}<0$. Capacitated: $\bar{c}_{ij}\ge 0\ \forall (i,j)\in D$ and $\bar{c}_{ij}\le 0\ \forall (i,j)\in U$. For selected $(i,j)$, if $(i,j)\in D$, forward is $i\to j$ and if $(i,j)\in U$, forward is $j\to i$. (4) Adding $(i,j)$ to $T$ creates a unique cycle. If $B=\emptyset\Longrightarrow\theta^*=\infty$ and terminate with optimal cost $-\infty$. Else choose $\theta^*$ defined above. (5) Push $\theta^*$ units of flow around the cycle and remove the arc whose flow hits a bound.

**Integrality.** $A$ integers $\Rightarrow B,B^{-1}$ integers. $B^{-1}$ integer since $B$ integer and lower triangular with diagonal entries $\in\{1,-1\}$. $b_i$ $(c_{ij})$ integers, $\Rightarrow$ basic solution (dual basic solution) integers

# 8 Ellipsoid and interior point methods

## 8.1 Ellipsoid method

**Algorithm. (Inputs)** $P=\{Ax\ge b\}$, $t^*=t(v)$ such that $Vol(P)<v\Rightarrow P=\emptyset$, $P\supset E_0=E(x_0,r^2I)$ **(Iteration)** (1) If $t=t^*$, $P=\emptyset$. If $x_t\in P$, $P\neq\emptyset$. Else, find $i$ such that $a_i^\top x_t<b_i$. (2) Let $H_t=\{x\mid a_i^\top x\ge a_i^\top x_t\}$, build $E_{t+1}\supseteq H_t\cap E_t$, set $x_{t+1}\leftarrow\text{center}(E_{t+1})$ **(Polynomial time)** Each cut reduces ellipsoid volume by a fixed multiplicative factor $e^{-1/(2(n+1))}$

**Relaxing requirement on boundedness.** All extreme points $x^{(i)}$ of $P$ are contained in a bounded polyhedron, $P_B$. If $\text{rank}(A)=n$, $P$ nonempty $\Longrightarrow P$ has at least one extreme point and the algorithm can be applied to $P_B$.



**Relaxing requirement on empty or full-dimensional.** Define $P_e=\{x\mid Ax\ge b-\epsilon\mathbf{1}\}$ for special $\epsilon$ and show $P$ empty (nonempty) $\Rightarrow P_e$ empty (full-dimensional) and the algorithm can be applied to $P_e$.

**For linear programs.** Find feasible $(x,p)\in\mathcal{O}=\{(x,p)\in\mathbb{R}^{n+m}\mid Ax=b,\ x\ge 0,\ c^\top x=p^\top b, p^\top A\le c^\top\}$. Can check feasibility of $P=\{x\mid Ax\ge b\}$ for primal feasibility. Can check feasibility of $P=\{d\mid Ad\ge 0, c^\top d=-1\}$ for unboundedness. **(Bisection)** To solve $\min_{x\in P} c^\top x$, maintain an interval $[L,U]$ known to contain the optimal value, $c^\top x^*$. At each iteration, query feasibility of $P(p)=\{x\mid Ax\ge b,\ c^\top x\le p\}$ at midpoint $p=(L+U)/2$. If P(p) is feasible, set $U\leftarrow p$, else set $L\leftarrow p$.

**Separation problem.** If we can solve a separation problem (decide $x\in P$ or produce $d$ such that $d^\top x<d^\top y\ \forall\ y\in P$) for $P$ polynomial time, we can solve an optimization problem over $P$ in polynomial time. The converse is also true: Define $P^o=\{y\mid y^\top x\le 1\ \forall\ x\in P\}$ and assume $0\in P$. For $OPT(P)=\max_{x\in P} c^\top x$, (i) if $c^\top x^*\le 1\Longrightarrow c^\top x\le 1\ \forall x\in P\Longrightarrow c\in P^o$. (ii) if $c^\top x^*>1\Longrightarrow\exists x^o\in P$ such that $c^\top x^o>1$ and note $z^\top x^o\le 1\ \forall z\in P^o$ so $x^o$ separates $c$ and $P^o$. Finally, $P=P^{oo}$, so $OPT(P)\iff SEP(P^o)\iff OPT(P^o)\iff SEP(P^{oo})=SEP(P)$. (Can also use Farka's lemma for separating hyperplane)

**E.g., Traveling salesman.** Separation problem for a given $\tilde{x}$ is the min-cut problem with $u_e=\tilde{x}_e$ and $\mathcal{A}=\{\mathcal{E}_\rightarrow,\mathcal{E}_\leftarrow\}$. Let $S_0$ be the minimum cut. Then if $\sum_{e\in\delta(S_0)} x_e=\sum_{e\in\mathcal{E}}\tilde{x}_ez_e<2$, $\tilde{x}_e$ infeasible and $S_0$ is the violated constraint.

$$\min_{x_e}\sum_{e\in\mathcal{E}} c_ex_e \qquad \text{s.t. } \sum_{e\in\delta(\{i\})} x_e=2\ \forall\, i\in\mathcal{N},\ x_e\in[0,1]\ \forall\, e\in\mathcal{E},\ \sum_{e\in\delta(S)} x_e\ge 2\ \forall\, S\subset\mathcal{N}\setminus\{\emptyset\} \qquad (\delta(S)=\{(i,j)\mid i\in S,\ j\notin S\})$$

**E.g., Min-weight edge set to intersect every $s\to t$ path.** Separation problem for a given $\tilde{x}$ is the shortest path problem with $l_{ij}=\tilde{x}_{ij}$ and $\mathcal{A}=\{\mathcal{E}_\rightarrow,\mathcal{E}_\leftarrow\}$. Let $K_0$ be the shortest path. Then if $\sum_{e\in K_0}\tilde{x}_e<1$, $\tilde{x}_e$ infeasible and $K_0$ is the violated constraint.

$$\min_{x_e}\sum_{e\in\mathcal{E}} c_ex_e \qquad \text{s.t. } \sum_{e\in K} x_e\ge 1\ \forall\ K\in\mathcal{K},\ 0\le x_e\le 1,\ \forall\, e\in\mathcal{E} \qquad (\mathcal{K}\text{ is the set of all } s\to t \text{ paths})$$

## 8.2 Affine scaling

**Approach** For $P=\{x\mid Ax=b,\ x\ge 0\}$, stay strictly in the interior of $P$ : $\text{relint}(P)=\{x\mid Ax=b,\ x>0\}$

**Dikin ellipsoid and local LP** At $x_k>0$, the Dikin ellipsoid, $E_k=\{x\mid (x-x_k)^\top X_k^{-2}(x-x_k)\le 1\}$, $X_k=\text{diag}(x_k)$, is the largest ellipsoid centered at $x_k$ satisfying $E_k\subseteq\text{relint}(P)$. Affine scaling computes the next iterate as the minimizer of the linear objective over this ellipsoid: $x_{k+1}=\arg\min\{c^\top x\mid x\in E_k\cap\{x\mid Ax=b\}\}$. Since the boundary of $E_k$ is strictly interior to $P$, $x_{k+1}$ remains feasible and strictly positive.

**Direction computation.** For $d=x^{k+1}-x^k$, $x^{k+1}\in P$, we equivalently solve $\min_{\{d\mid Ad=0,\ d^\top X_k^{-1}d\le 1\}} c^\top d$, which has an analytical solution. Further, $c^\top d=c^\top(x^{k+1}-x^k)<0$ so objective strictly improves. **(Interpretation)** $d_k=x_{k+1}-x_k$ is the direction of steepest feasible descent after scaling the geometry of $P$ by the current point. The scaling $X_k^{-1}$ normalizes coordinates so the step is "round" in the interior; this is why $E_k$ always fits inside relint$(P)$.

**Algorithm (Iteration)** (i) Compute $X_k=\text{diag}(x^k)$, $p^k=(AX_k^2A^\top)^{-1}AX_k^2c$, $r^k=c-A^\top p^k$. (ii) if $r^k\ge 0$, $x^{k\top}r^k<\epsilon$, then $\epsilon$-optimal $x^k$ found. (iii) If $X_k^2r^k\ge 0$, then optimal cost is unbounded. (iv) $x^{k+1}=x^k-\beta X_k^2r^k/||X_kr^k||_2$. Longstep variants update $||X_kr^k||_2$ with $||X_kr^k||_\infty$ or $\max_{\{X_{ki}r_i^k>0\}} X_{ki}r_i^k$. **(Initialize)** Want $x_0>0$. Can be done with $\min_{\{Ax+(b-A\mathbf{1})x_{n+1}=b,\ (x,x_{n+1})\ge 0\}} c^\top x+Mx_{n+1}$ and starting point $(\mathbf{1},1)$.

## 8.3 Barrier method and central path

**Setup.** Exam questions will focus on (i) taking derivatives of the barrier Lagrangian and (ii) using $x^*\odot s^*=\mu$. Affine scaling moves quickly inside $P$ but slows near the boundary. Barrier methods instead pick a path that stays strictly interior for all $\mu>0$. Consider primal-dual pair $(P):\min_{\{x\mid Ax=b,\ x\ge 0\}} c^\top x$, $(D):\max_{\{p\mid p^\top A+s=c^\top,\ s\ge 0\}} p^\top b$. Note the duality gap identity $c^\top x-p^\top b=(s^\top+p^\top A)x-p^\top Ax=s^\top x$. To implement, consider expressing one variable in terms of the other ones (e.g., $a_2x_2=b-a_1x_1$)

**Barrier problems.** Introduce logarithmic barriers on the nonnegativity constraints:

$$\min_{\{x\mid Ax=b\}} B_\mu(x)=c^\top x-\mu\sum_{i=1}^{n}\log x_i \qquad \max_{\{p,s\mid p^\top A+s=c^\top\}} B_\mu(x)=p^\top b-\mu\sum_{j=1}^{n}\log s_j$$

For the barrier problems, the necessary and sufficient (KKT) conditions for optimality are

$$Ax(\mu)=b \qquad x(\mu)\ge 0 \qquad A^\top p(\mu)+s(\mu)=c \qquad s(\mu)\ge 0 \qquad x(\mu)\odot s(\mu)=\mu$$

Any $(x(\mu),\ p(\mu),\ s(\mu))$ satisfying these is optimal for the barrier problems. To check optimality, examine Lagrangian:

$$L(x,p^*)=c^\top x-p^{*\top}(Ax-b)-\mu\sum_i\log x_i=(A^\top p^*+s^*)^\top x-p^{*\top}(Ax-b)-\mu\sum_i\log x_i=\sum_{i=1}^{n}(s_i^*x_i-\mu\log x_i)-p^{*\top}b \quad \left(\tfrac{\partial L(x,p^*)}{\partial x_i}=0\iff x_i=\tfrac{\mu}{s_i^*}\right)$$

So minimizing the Lagrangian gives minimizer $x^*=x(\mu)=\mu/s^*$

**Central path.** The solutions $(x(\mu),s(\mu))$ parameterized by $\mu>0$ are the **central path**. As $\mu\to 0$, $x(\mu)\to x^*$ and $s(\mu)\to s^*$ for some optimal primal-dual pair.

**Exercise.** Let $(x_\mu,p_\mu)$ a saddle point of $L(x,p)=c^\top x+p^\top b-p^\top Ax-\mu\sum_i\log x_i$. Show $(x_\mu,s_\mu=c-A^\top p_\mu)$ lies on central path:

$\frac{\partial L(x,p)}{\partial x}=0\iff c_i-p^\top A_i-\frac{\mu}{x_i}=0\ \forall\, i.$ Let $s_i:=c_i-p^\top A_i$ then $s_ix_i=\mu\ \forall\, i$ and $s_i>0\ \forall\, i$ since $\mu,x_i>0$

$\frac{\partial L(x,p)}{\partial p}=0\iff b-Ax=0\iff Ax=b$ Together we have $\{x>0,\ s>0,\ p^\top A+s^\top=c^\top,\ Ax=b,\ s\odot x=\mu\}$ (central path of standard form primal)

**Exercise.** Suppose for $\mu=1$ the point $x_\mu$ is feasible but not optimal, yet $(x_\mu,s_\mu)$ converge as $\mu\to 0$. Show (i) some index $i$ satisfies $x_i^*=0$, (ii) for that $i$, $\lim_{\mu\to 0}x_{\mu,i}/\mu>0$. **(Proof i)** If $x_i^*>0\ \forall\, i$, then strict complementarity gives $s_i^*=0\ \forall\, i$. Then any feasible $x$ satisfies $x\odot s^*=0$, so all feasible $x$ are optimal — a contradiction to $x_\mu$ feasible but not optimal. **(Proof ii)** For the index $i$ with $x_i^*=0$, strict complementarity gives $s_i^*>0$. From $x_{\mu,i}s_{\mu,i}=\mu$, $\frac{x_{\mu,i}}{\mu}=\frac{1}{s_{\mu,i}}\to\frac{1}{s_i^*}>0$.